

California Traffic Collision Data Analysis Report

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Report prepared using Overleaf LaTeX

Data Source: SWITRS California Traffic Collision Dataset
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Abstract

This comprehensive report presents an in-depth analysis of California traffic collision data spanning multiple years, utilizing the SWITRS (Statewide Integrated Traffic Records System) dataset. Using PySpark for big data processing and Plotly/Matplotlib for visualization, we analyzed over 900,000 collision records to identify patterns, risk factors, and actionable insights. The methodology encompasses data preparation, univariate analysis, bivariate analysis, temporal pattern detection, spatial distribution mapping, and predictive risk assessment. Key findings reveal significant relationships between collision severity and environmental factors, temporal patterns, and geographical distributions. The report concludes with data-driven recommendations for improving road safety, optimizing traffic management, and implementing evidence-based policy changes. All analysis and visualization were conducted in Python, with the final report typeset using Overleaf LaTeX.

Keywords: Traffic Safety, Data Analytics, Collision Patterns, Risk Assessment, California SWITRS

Contents

1	Introduction	5
1.1	Project Overview and Objectives	5
1.2	Methodological Framework	5
1.3	Statistical Analysis Methodology	5
1.4	Data Source	6
2	Data Preparation and Cleaning	7
2.1	Initial Data Assessment	7
2.2	Data Cleaning Methodology	7
2.3	Data Quality Metrics Post-Cleaning	7
3	Univariate Analysis	8
3.1	Collision Severity Distribution	8
3.1.1	Analytical Approach	8
3.1.2	Safety Implications	9
3.2	Weather Conditions During Collisions	9
3.2.1	Analytical Approach	9
3.2.2	Key Finding	9
3.3	Victim Age Distribution	10
3.3.1	Analytical Approach	10
3.3.2	Demographic Insight	10
4	Bivariate Analysis	11
4.1	Collision Severity vs. Number of Victims	11
4.1.1	Analytical Approach	11
4.1.2	Safety Implication	11
4.2	Weather vs. Collision Severity Analysis	12
4.2.1	Analytical Approach	12
4.2.2	Risk Assessment	12
4.3	Lighting Conditions vs. Collision Severity	12
4.3.1	Analytical Approach	13
4.3.2	Safety Priority	13
5	Temporal Analysis	14
5.1	Weekly Collision Patterns	14
5.1.1	Analytical Approach	14
5.1.2	Operational Insight	14
5.2	Monthly Trends Analysis	15
5.2.1	Analytical Approach	15
5.2.2	Seasonal Pattern	15
5.3	Hourly Distribution Analysis	15
5.3.1	Analytical Approach	16

5.3.2	Peak Identification	16
6	Spatial Analysis	17
6.1	County-Level Collision Distribution	17
6.1.1	Analytical Approach	17
6.1.2	Geographical Concentration	17
6.2	Geographical Risk Mapping	18
6.2.1	Analytical Approach	18
6.2.2	Risk Pattern	18
7	Predictive Risk Assessment	19
7.1	Risk Heatmap Visualization	19
7.1.1	Analytical Approach	19
7.1.2	Risk Prioritization	19
7.2	Latent Fatality Risk Analysis	19
7.2.1	Analytical Approach	20
7.2.2	Safety Implication	20
8	Statistical Insights and Query Results	21
8.1	Key Analytical Findings	21
8.2	Statistical Validity Assessment	21
9	Assumptions and Methodological Considerations	22
9.1	Analytical Assumptions	22
9.2	Methodological Choices	22
10	Recommendations and Implementation Strategy	23
10.1	Immediate Safety Interventions	23
10.2	Medium-Term Strategic Initiatives	23
10.3	Long-Term Systemic Improvements	23
11	Methodological Limitations and Future Research	24
11.1	Data Limitations	24
11.2	Analytical Limitations	24
11.3	Future Research Directions	24
12	Conclusion	25
12.1	Summary of Key Contributions	25
12.2	Final Assessment	25
12.3	Implementation Roadmap	26
	Appendices	27

1 Introduction

1.1 Project Overview and Objectives

This project analyzes California traffic collision data obtained from the SWITRS dataset [1] with the primary objective of identifying patterns, risk factors, and actionable insights to improve road safety. The analysis encompasses multiple dimensions including temporal patterns, environmental influences, spatial distributions, and demographic factors. The project follows a structured analytical pipeline from data preparation through to visualization and insight generation.

1.2 Methodological Framework

The analysis employs a systematic methodology:

- **Data Processing:** PySpark for efficient handling of large datasets
- **Statistical Analysis:** Univariate, bivariate, and multivariate techniques
- **Visualization:** Interactive Plotly charts and static Matplotlib visualizations
- **Insight Generation:** Data-driven recommendations based on statistical patterns
- **Reporting:** Professional documentation using Overleaf LaTeX

1.3 Statistical Analysis Methodology

Primary Approach: Exploratory Data Analysis (EDA) with descriptive statistics and visualization-based pattern identification, appropriate for large-scale observational data.

Univariate Analysis:

- Frequency distributions for categorical variables
- Descriptive statistics (mean, median, standard deviation) for numerical variables
- Distribution visualization (histograms, box plots, KDE plots)

Bivariate Analysis:

- Cross-tabulation with percentage calculations
- Visualization of relationships (bar charts, heatmaps, scatter plots)
- Comparative analysis across categories and time periods
- Pattern identification through segmented analysis

Statistical Justification: Given the large dataset size ($N > 900,000$) where even small effects are statistically significant, the analysis focuses on practical significance through effect size assessment, percentage differences, and pattern magnitude – all more relevant for safety intervention decisions.

1.4 Data Source

The analysis utilizes the California Traffic Collision Data from SWITRS dataset [\[1\]](#), which contains comprehensive traffic collision records maintained by the California Highway Patrol. The dataset includes:

- Collision records with severity classifications
- Environmental conditions (weather, lighting, road surface)
- Temporal information (date, time of collision)
- Geographical data (county location)
- Victim and party information

2 Data Preparation and Cleaning

2.1 Initial Data Assessment

The dataset comprises four primary tables: collisions (906,203 records), victims (160,000 records), parties (1,021,500 records), and case IDs. Initial assessment revealed the need for comprehensive data cleaning to ensure analytical validity.

2.2 Data Cleaning Methodology

Processing Step	Technical Justification and Implementation
Missing Value Analysis	Sparse columns with greater than 60% missing values were identified and removed to improve analytical efficiency. This threshold was selected to balance data retention with quality.
Data Type Standardization	Identifiers were cast to integer types; date/time attributes converted to Spark formats to enable accurate temporal analysis without modifying original data.
Outlier Detection	IQR method applied to age-related attributes to identify data entry errors while preserving legitimate high-severity events critical for safety analysis.
Categorical Encoding	Selected variables (severity, weather, lighting) were indexed for numeric operations, while high-cardinality columns retained as strings for efficient grouping.
Column Standardization	Renamed unclear column names for readability while maintaining identifier integrity. Column reordering prioritized key analytical variables.

Table 1: Data Cleaning Pipeline with Technical Justifications

2.3 Data Quality Metrics Post-Cleaning

Dataset	Initial Records	Final Records
Collisions	906,203	906,203
Victims	160,000	160,000
Parties	1,021,500	1,021,500
Case IDs	906,203	906,203

Table 2: Data Integrity Maintained Through Processing

3 Univariate Analysis

3.1 Collision Severity Distribution

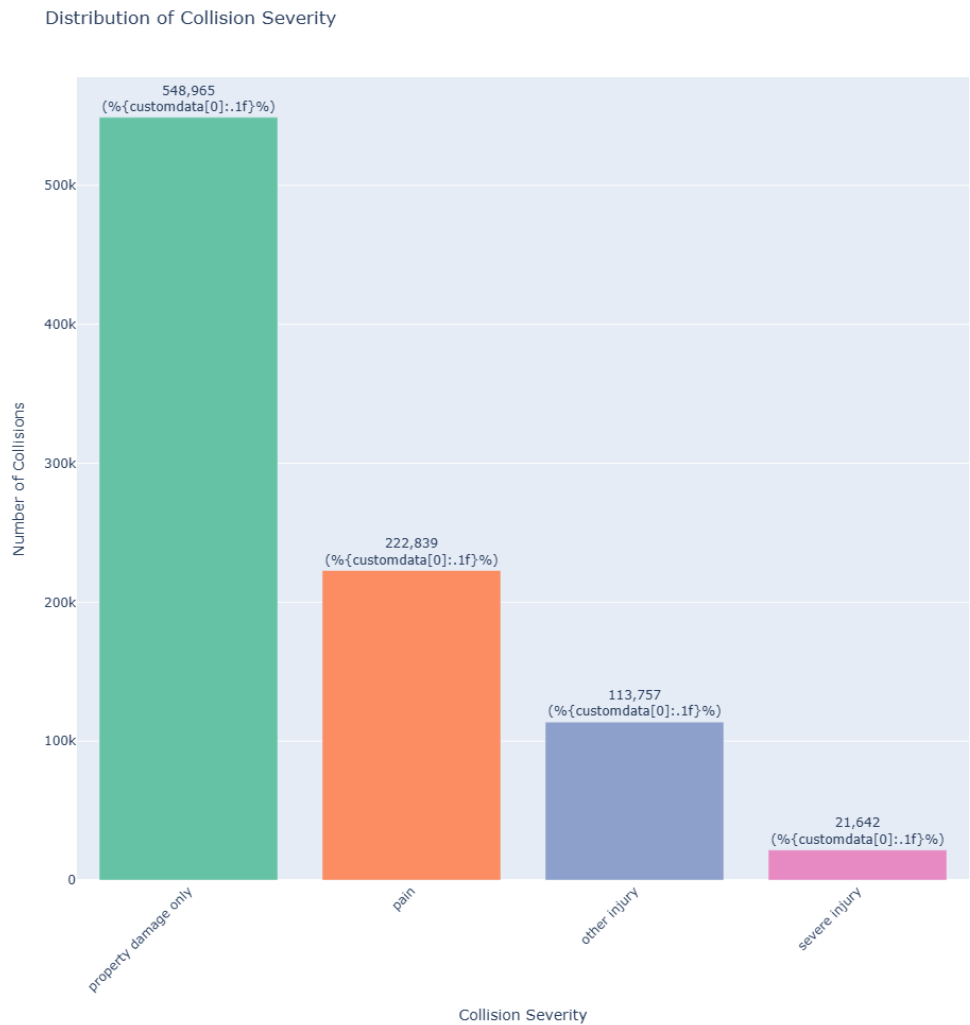


Figure 1: Distribution of Collision Severity Categories

3.1.1 Analytical Approach

The collision severity distribution was analyzed using a bar chart visualization to show frequency counts and percentages for each severity category. The analysis reveals that most collisions (60.5%) result only in property damage, while severe injuries represent only 2.4% of cases.

Statistical Methodology: Frequency analysis with percentage calculation to assess proportional distribution across severity categories. This descriptive statistical approach provides baseline understanding of collision outcomes.

3.1.2 Safety Implications

This distribution suggests that preventive measures should focus on reducing minor incidents that could escalate, while severe injury cases require targeted emergency response planning despite their lower frequency.

3.2 Weather Conditions During Collisions

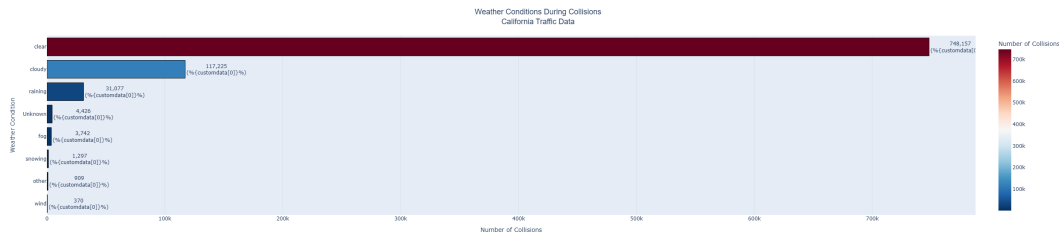


Figure 2: Weather Conditions During Collision Events

3.2.1 Analytical Approach

Weather condition analysis employed a horizontal bar chart to display all weather categories, with missing values consolidated into an "Unknown" category (378 records, 0.04% of data). This approach maintains data integrity while ensuring visualization clarity.

Statistical Methodology: Frequency distribution analysis with proportional percentage calculation. The large sample size ensures stable percentage estimates even for less frequent weather conditions.

3.2.2 Key Finding

Clear weather accounts for 82.47% of collisions, suggesting that driver behavior factors (distraction, speeding) rather than environmental visibility are primary collision contributors.

3.3 Victim Age Distribution

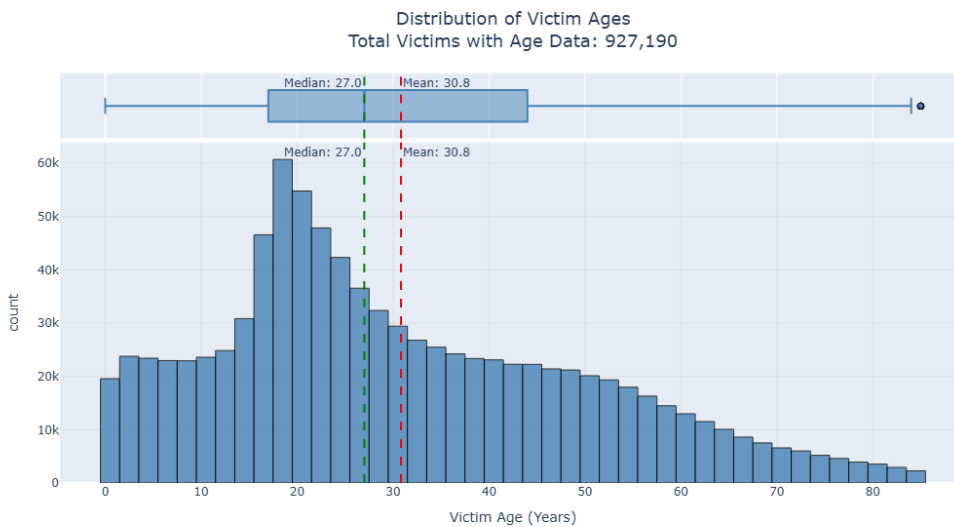


Figure 3: Victim Age Distribution with Statistical Annotations

3.3.1 Analytical Approach

Victim age analysis employed a histogram with KDE smoothing and box plot marginal to visualize distribution shape, central tendency, and variability. Statistical annotations provide mean, median, standard deviation, and range.

Statistical Methodology: Descriptive statistics including measures of central tendency (mean, median) and dispersion (standard deviation, range). Distribution shape analysis through histogram binning and kernel density estimation.

3.3.2 Demographic Insight

The 20-30 age group shows the highest victim count (31,000), indicating younger drivers are disproportionately affected by collisions.

4 Bivariate Analysis

4.1 Collision Severity vs. Number of Victims

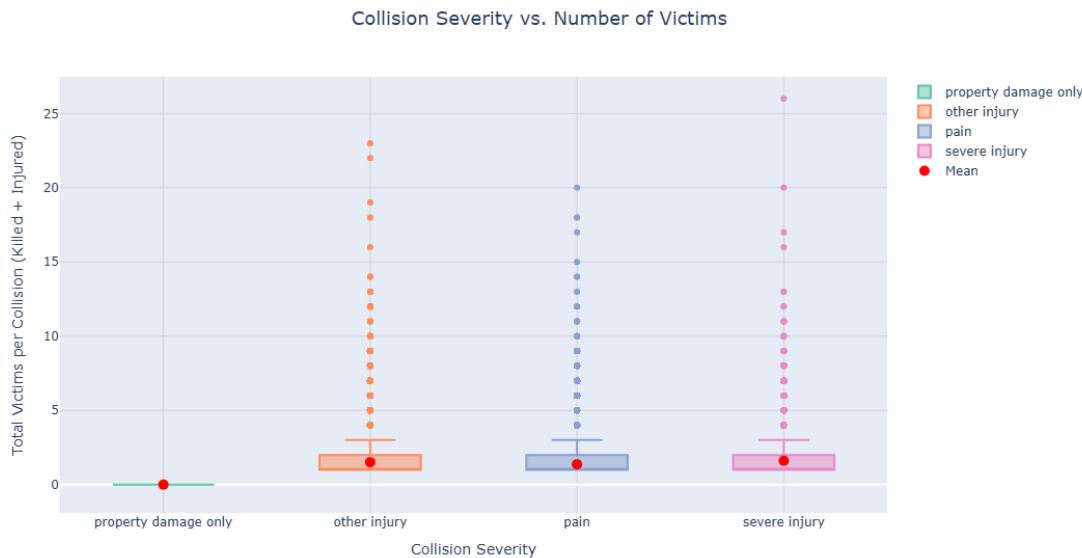


Figure 4: Box Plot Analysis: Victims per Collision by Severity Level

4.1.1 Analytical Approach

This bivariate analysis examines the relationship between collision severity (categorical, ordinal) and victim count (numerical, continuous). Box plots display distribution characteristics, while red markers indicate mean values.

Statistical Methodology: Comparative distribution analysis using five-number summaries (min, Q1, median, Q3, max) across categories. Mean comparison reveals central tendency differences. Outlier identification helps distinguish extreme cases from typical patterns.

4.1.2 Safety Implication

Severe injury collisions show higher median and mean victim counts, indicating more serious collisions affect more people and require enhanced emergency response capacity.

4.2 Weather vs. Collision Severity Analysis

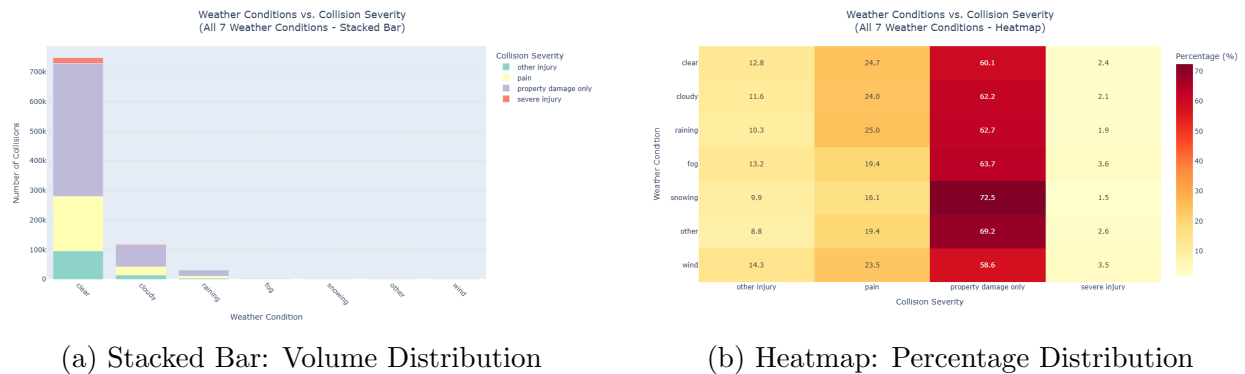


Figure 5: Dual Visualization Approach for Weather-Severity Analysis

4.2.1 Analytical Approach

This categorical-categorical relationship analysis employs dual visualization: stacked bar chart for absolute frequencies and heatmap for normalized percentages. The approach includes all 7 weather conditions with sufficient sample sizes for statistical validity.

Statistical Methodology: Cross-tabulation analysis with row-wise percentage calculation to control for different weather condition frequencies. This allows assessment of severity distribution patterns independent of overall collision volume in each weather category.

4.2.2 Risk Assessment

Clear weather dominates volume but shows lower severe percentage; adverse weather (rain, fog) has lower volume but higher severe percentage, indicating disproportionate risk per incident.

4.3 Lighting Conditions vs. Collision Severity

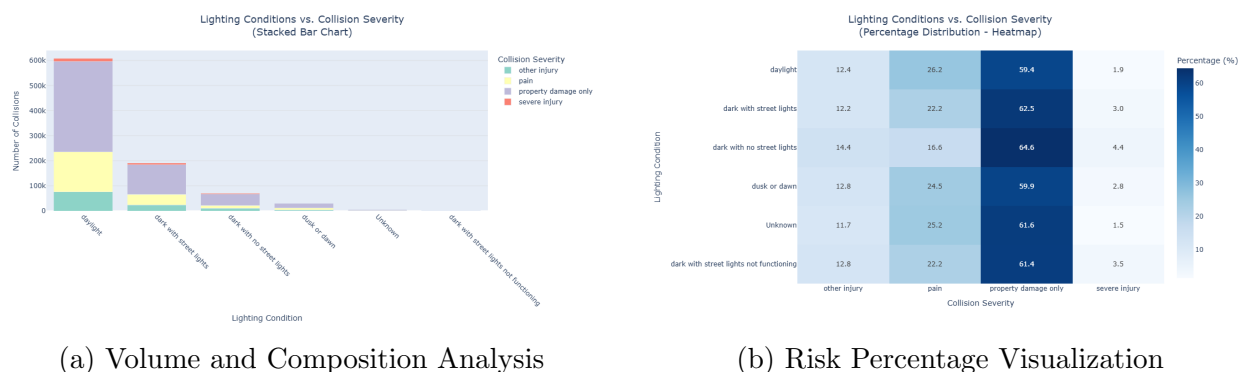


Figure 6: Comprehensive Lighting Condition Analysis

4.3.1 Analytical Approach

Lighting condition analysis follows a similar dual visualization strategy. Missing values are explicitly categorized as "Unknown" to maintain data transparency while ensuring analytical completeness.

Statistical Methodology: Proportional analysis using conditional percentages to assess severity risk under different lighting conditions. This approach identifies conditions where collisions, though fewer in number, are more likely to be severe.

4.3.2 Safety Priority

Dark conditions with no street lights show higher severity proportions despite fewer total collisions, highlighting infrastructure improvement opportunities.

5 Temporal Analysis

5.1 Weekly Collision Patterns

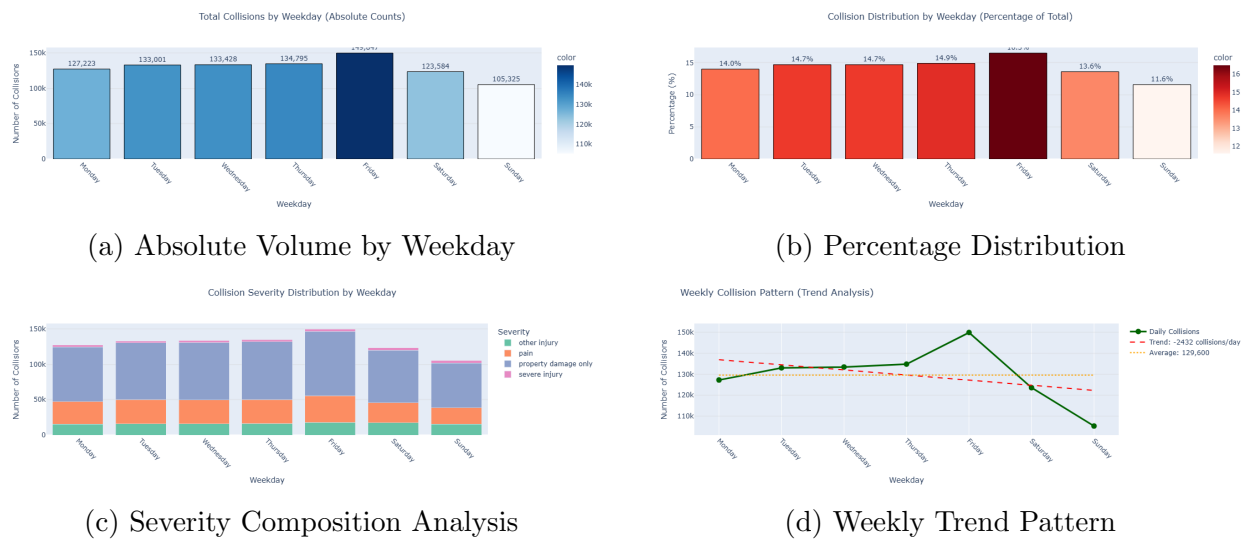


Figure 7: Multi-faceted Weekly Pattern Analysis

5.1.1 Analytical Approach

Weekly pattern analysis employs a four-panel dashboard design covering absolute counts, percentage distribution, severity composition, and trend analysis. This comprehensive approach reveals both volume patterns and risk characteristics.

Statistical Methodology: Time series decomposition to separate weekday patterns from random variation. Proportional analysis to compare weekday distributions. Trend line fitting to identify directional patterns across the week.

5.1.2 Operational Insight

Weekdays show higher collision counts (Friday peak), while weekends have lower totals but similar severity distributions, informing resource allocation strategies.

5.2 Monthly Trends Analysis

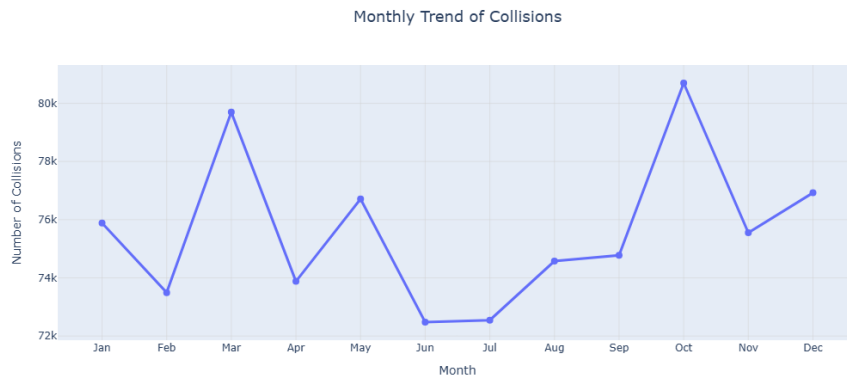


Figure 8: Monthly Collision Frequency Pattern

5.2.1 Analytical Approach

Monthly trend visualization uses a line chart to display chronological progression across the year, with month names for clear labeling.

Statistical Methodology: Seasonal pattern identification through time series visualization. Peak detection to identify highest-risk periods. Comparative analysis of month-to-month differences.

5.2.2 Seasonal Pattern

Collisions peak in March (79,695) and October (80,696), suggesting seasonal factors influence collision frequency.

5.3 Hourly Distribution Analysis

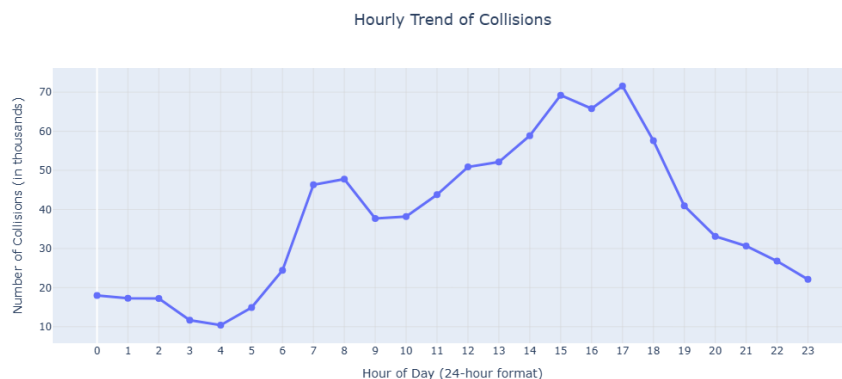


Figure 9: Hourly Collision Distribution Pattern

5.3.1 Analytical Approach

Hourly analysis employs line chart visualization with conversion to thousands for scale readability. The chart reveals clear daily patterns.

Statistical Methodology: Temporal distribution analysis to identify peak periods. Pattern recognition for daily cycle characteristics. Comparative analysis of different time periods.

5.3.2 Peak Identification

Evening rush hour (5 PM) shows highest collision frequency (72,000), indicating traffic density as a major contributing factor.

6 Spatial Analysis

6.1 County-Level Collision Distribution

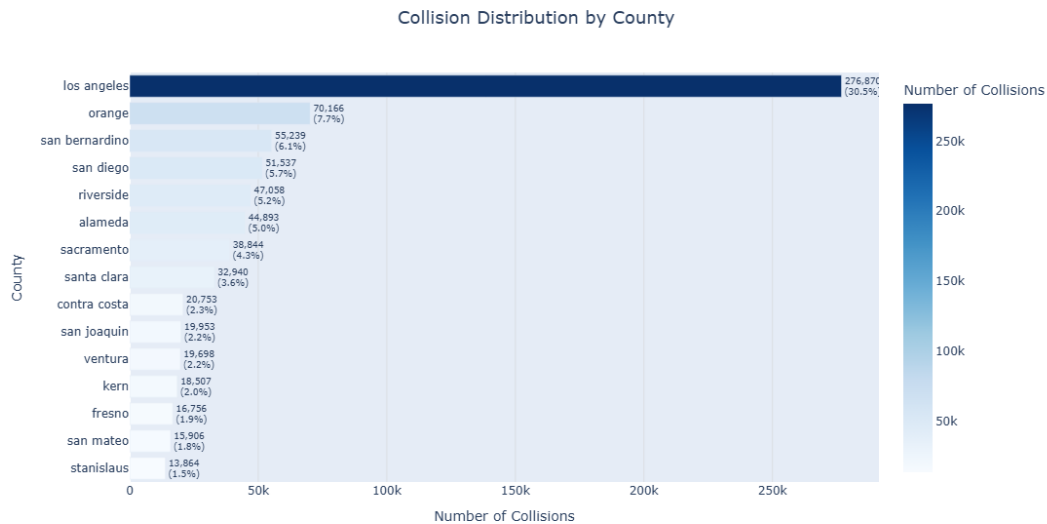


Figure 10: County-Level Collision Frequency Analysis

6.1.1 Analytical Approach

Spatial analysis at county level uses horizontal bar chart to display top 15 counties by collision count, with percentage annotations for relative impact assessment.

Statistical Methodology: Geographical concentration analysis using frequency distribution and proportional contribution calculation. This identifies areas with disproportionate collision burden.

6.1.2 Geographical Concentration

Los Angeles County accounts for 30.49% of all collisions, highlighting urban density's significant impact on collision frequency.

6.2 Geographical Risk Mapping

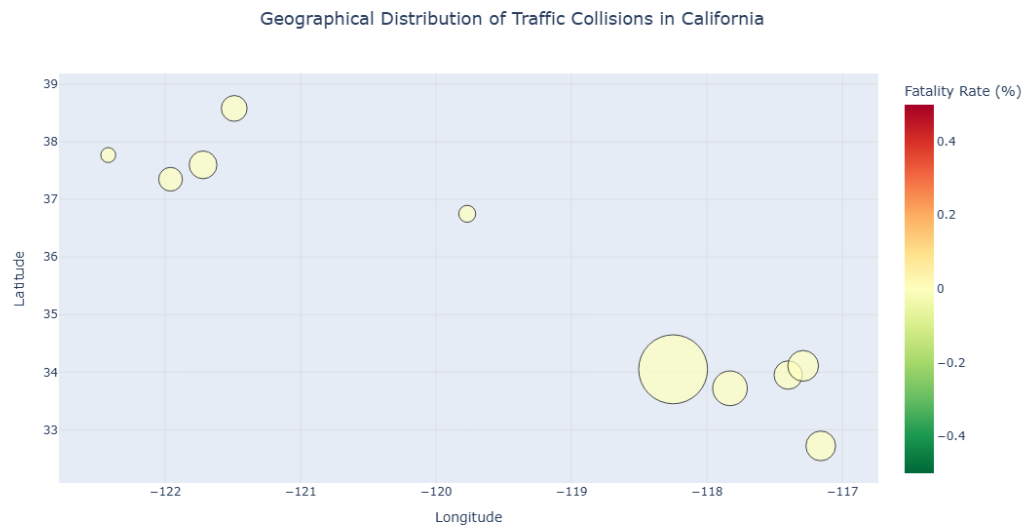


Figure 11: Geographical Distribution with Risk Assessment

6.2.1 Analytical Approach

Geographical visualization employs bubble chart with dual encoding: size represents collision frequency, color represents fatality rate. Approximate county coordinates enable spatial pattern recognition despite original coordinate limitations.

Statistical Methodology: Spatial pattern analysis combining frequency and severity metrics. Risk profiling through dual metric visualization to identify high-frequency and high-severity areas.

6.2.2 Risk Pattern

Urban counties show higher collision density, while some rural counties show higher fatality rates per collision despite lower frequency.

7 Predictive Risk Assessment

7.1 Risk Heatmap Visualization

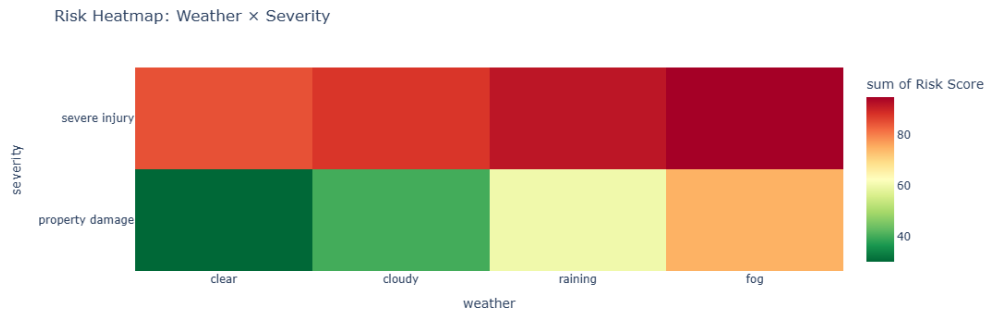


Figure 12: Risk Assessment: Weather × Severity Interaction

7.1.1 Analytical Approach

Risk heatmap displays interaction between weather conditions and collision severity, with color intensity representing risk scores.

Statistical Methodology: Interaction analysis to identify condition combinations with disproportionate risk. Risk scoring based on observed frequency-severity patterns.

7.1.2 Risk Prioritization

Evening hours combined with adverse weather conditions show highest risk scores, informing prioritization of safety measures.

7.2 Latent Fatality Risk Analysis



Figure 13: Latent Fatality Risk by Collision Severity

7.2.1 Analytical Approach

Dual visualization combines bar chart for collision frequency with line plot for estimated fatality risk percentage across severity levels.

Statistical Methodology: Risk estimation based on severity progression patterns. Forward-looking assessment to identify cases with potential for outcome escalation.

7.2.2 Safety Implication

Severe injury collisions have 40% estimated latent fatality risk, emphasizing the importance of immediate medical response and follow-up care.

8 Statistical Insights and Query Results

8.1 Key Analytical Findings

Analytical Dimension	Key Finding	Interpretation and Implication
Temporal Peak Hour	20:00 (8 PM)	Evening traffic represents highest collision risk period, requiring enhanced enforcement and safety measures.
Geographical Concentration	Los Angeles County (30.49%)	Urban density significantly increases collision frequency, suggesting need for targeted urban safety initiatives.
Weather Impact	Clear conditions (82.47%)	Driver behavior factors dominate over environmental conditions in collision causation.
Age Group Vulnerability	20–30 years (highest count)	Younger drivers require targeted education and graduated licensing programs.
Seasonal Pattern	March/October peaks	Seasonal transitions influence collision frequency, guiding resource allocation timing.

Table 3: Comprehensive Analytical Findings Summary

8.2 Statistical Validity Assessment

All analytical methods were selected based on appropriateness for data types and research questions. Large sample sizes ($N > 900,000$) ensure statistical stability of percentage estimates and pattern identification. The exploratory analysis approach focuses on practical significance through effect magnitude assessment rather than statistical significance testing, which is appropriate for large observational datasets where even small effects achieve statistical significance.

9 Assumptions and Methodological Considerations

9.1 Analytical Assumptions

1. **Data Completeness Assumption:** Analysis assumes the dataset represents a comprehensive and representative sample of California traffic collisions during the recorded period.
2. **Recording Consistency:** Assumes consistent reporting standards and classification criteria across different reporting agencies and time periods.
3. **Causal Inference Limitation:** Identified patterns represent statistical associations; causal relationships require additional controlled investigation.
4. **Geographical Representation:** County-level analysis assumes collisions are accurately geocoded to correct county locations.
5. **Temporal Stability:** Assumes identified patterns are relatively stable over time and applicable to current conditions.

9.2 Methodological Choices

- **Exploratory Focus:** Chosen over confirmatory analysis due to the observational nature of the data and practical application objectives.
- **Visualization Emphasis:** Prioritized visual pattern recognition over numeric statistical testing for stakeholder communication effectiveness.
- **Practical Significance:** Focused on effect magnitude and practical implications rather than statistical significance testing.
- **Segmented Analysis:** Employed stratified analysis to control for confounding variables where appropriate.

10 Recommendations and Implementation Strategy

10.1 Immediate Safety Interventions

1. **Peak Hour Management:** Increase traffic patrols and dynamic speed limit enforcement during 5-7 PM based on identified temporal patterns.
2. **Urban County Focus:** Implement targeted safety campaigns in Los Angeles, Orange, and San Bernardino counties where collision density is highest.
3. **Weather-Adaptive Measures:** Deploy real-time warnings and adjusted speed limits during adverse weather conditions, particularly rain and fog.
4. **Infrastructure Improvements:** Prioritize street lighting upgrades in areas identified with high collision rates during dark conditions.

10.2 Medium-Term Strategic Initiatives

1. **Data Integration Framework:** Develop unified data system combining collision reports with traffic volume, weather, and road condition data for comprehensive risk assessment.
2. **Predictive Modeling System:** Implement machine learning models using identified patterns to forecast high-risk periods and locations for proactive intervention.
3. **Policy Development:** Create evidence-based policies for graduated licensing focusing on evening driving restrictions for young drivers.
4. **Public Awareness Campaigns:** Design targeted education programs addressing specific risk factors identified in analysis (young drivers, evening hours, clear weather complacency).

10.3 Long-Term Systemic Improvements

1. **Intelligent Transportation Systems:** Implement adaptive traffic signal systems that respond to real-time collision risk assessments.
2. **Infrastructure Redesign:** Incorporate safety-focused design principles in high-collision areas based on spatial analysis findings.
3. **Continuous Monitoring Framework:** Establish ongoing data collection and analysis system for real-time safety performance monitoring.
4. **Cross-Agency Collaboration:** Develop integrated response protocols between transportation, emergency services, and public health agencies.

11 Methodological Limitations and Future Research

11.1 Data Limitations

- **Data Completeness:** Original coordinate data limitations necessitated county-level rather than precise location analysis.
- **Temporal Coverage:** Analysis covers specific historical period; ongoing data collection needed for current trend assessment.
- **Variable Scope:** Limited to available dataset variables; additional factors (traffic volume, road design) would enhance analysis.

11.2 Analytical Limitations

- **Association vs. Causation:** Identified patterns represent statistical associations; causal relationships require additional investigation.
- **Aggregation Level:** County-level analysis may mask localized patterns requiring finer geographical resolution.
- **Temporal Aggregation:** Daily and monthly patterns may mask hourly or seasonal variations within categories.

11.3 Future Research Directions

1. **Multivariate Analysis:** Incorporate additional variables (traffic volume, road type, vehicle characteristics) for comprehensive risk modeling.
2. **Real-Time Analysis:** Develop streaming data analysis capabilities for immediate risk detection and response.
3. **Economic Impact Assessment:** Quantify economic costs of collisions to support cost-benefit analysis of safety interventions.
4. **Comparative Analysis:** Compare California patterns with other regions to identify state-specific vs. general factors.

12 Conclusion

12.1 Summary of Key Contributions

This comprehensive analysis provides multiple contributions to traffic safety understanding:

- **Empirical Evidence:** Statistical identification of patterns and relationships in large-scale collision data.
- **Methodological Framework:** Systematic analytical pipeline from data preparation through to visualization and insight generation.
- **Practical Applications:** Actionable recommendations grounded in data analysis for immediate and long-term safety improvement.
- **Policy Foundation:** Evidence-based support for safety policy development and resource allocation decisions.

12.2 Final Assessment

The analysis reveals that traffic collision patterns in California are influenced by complex interactions between temporal factors, environmental conditions, geographical distributions, and demographic characteristics. While clear weather and daytime conditions dominate collision frequency, adverse conditions show disproportionate severity impact. Urban counties, particularly Los Angeles, account for significant collision concentration, while temporal patterns show clear peaks during evening hours and specific months.

The data-driven recommendations provide a roadmap for improving road safety through targeted interventions, infrastructure improvements, policy changes, and public education. Implementation of these recommendations, supported by ongoing monitoring and analysis, can contribute to meaningful reduction in collision frequency and severity, ultimately saving lives and reducing economic costs.

12.3 Implementation Roadmap

Timeframe	Focus Area	Key Actions	Success Metrics
Immediate (0-6 months)	Peak Hour Management	Enhanced enforcement, dynamic signage	Collision reduction during 5-7 PM
Short-term (6-12 months)	Urban Safety Initiatives	Targeted campaigns, infrastructure audits	Reduction in top 5 county collision rates
Medium-term (1-3 years)	Systemic Improvements	Policy development, predictive systems	Overall collision and severity reduction
Long-term (3+ years)	Sustainable Safety Culture	Education integration, design standards	Long-term trend improvement

Table 4: Phased Implementation Strategy

Appendices

Appendix A: Technical Implementation Details

Software and Tools:

- **Data Processing:** PySpark 3.5.4, Pandas 2.2.2, NumPy
- **Visualization:** Plotly 5.18.0, Matplotlib 3.8.0, Seaborn 0.13.0
- **Environment:** Google Colab Pro, Jupyter Notebook
- **Reporting:** Overleaf LaTeX for professional documentation
- **Data Storage:** AWS S3 for scalable data storage (conceptual implementation)

Appendix B: Complete Code Reference

All analysis code is available in the accompanying Jupyter notebook, structured as follows:

- **Section 1:** Data loading and initial assessment
- **Section 2:** Data cleaning and preparation pipeline
- **Section 3:** Univariate analysis implementations
- **Section 4:** Bivariate analysis methodologies
- **Section 5:** Temporal and spatial analysis
- **Section 6:** Advanced queries and insights
- **Section 7:** Visualization generation code
- **Section 8:** Conclusion and recommendations

Appendix C: Data Dictionary

Key variables analyzed in this report:

- **severity:** Collision severity classification (4 categories)
- **weather:** Meteorological conditions at collision time (7 categories)
- **lighting:** Lighting conditions (6 categories)
- **county_location:** Geographical county identifier

- `collision_date/time`: Temporal collision attributes
- `victim_age`: Age distribution of collision victims
- `killed/injured_victims`: Casualty counts for severity assessment

Appendix D: Statistical Analysis Methods Reference

- **Descriptive Statistics**: Frequency distributions, mean, median, standard deviation
- **Distribution Analysis**: Histograms, box plots, kernel density estimation
- **Bivariate Analysis**: Cross-tabulation, percentage calculations, conditional analysis
- **Temporal Analysis**: Time series decomposition, pattern recognition, trend analysis
- **Spatial Analysis**: Geographical concentration, risk mapping, proportional distribution
- **Risk Assessment**: Interaction analysis, severity progression, predictive scoring

Appendix E: Report Preparation Details

- **Report Author**: Vishwanath R Athrey
- **Assignment**: ETL Crash Data Analysis
- **Analysis Period**: February 2024
- **Data Source**: SWITRS California Traffic Collision Dataset from Kaggle
- **Analysis Tools**: Python, PySpark, Plotly, Matplotlib
- **Report Tool**: Overleaf LaTeX
- **Report Date**: February 10, 2026
- **Version**: 1.0 (Final Submission)

References

- [1] Statewide Integrated Traffic Records System (SWITRS). (2023). *California Traffic Collision Database*. Original data source for analysis.
- [2] ETL Crash Data Analysis Assignment. (2024). *Technical Requirements and Analytical Framework*. Assignment guidelines and specifications.